Multi-D(n) Diophantine triples

Problèmes ouverts — A. Dujella · 2026

Problem 5.12 (Dujella, 2026). Are there infinitely many Diophantine triples that are also $D(n_i)$-triples for $1 < n_2 < n_3 < n_4$?

Remark. The following triples with the given property were found by Adžaga, Dujella, Kreso and Tadić (2018) : $\{4, 12, 420\}$, $\{10, 44, 21252\}$, $\{4, 420, 14280\}$, $\{40, 60, 19404\}$, $\{78, 308, 7304220\}$, $\{4, 485112, 16479540\}$, $\{15, 528, 32760\}$.